

Introduction to Congestion in Sensor Networks

Lecturer: Congestion and flow control in sensor networks. This topic is divided into two parts. Overall, we will cover how congestion is a serious problem in sensor networks, and the sensor network specific issues that must be taken into account while dealing with congestion. Then we will look into solutions proposed by researchers to handle congestion and flow control. Network congestion is undesirable in any network because it leads to wastage of communication and energy resources, and seriously affects data delivery reliability. This is due to congestion leading to packet drops.

WSN-Specific Congestion Sources

Lecturer: Packet drops decrease reliability of packet delivery to the sink or intended destination. In wireless sensor networks, unique sources of congestion include wireless shared medium, multi-hop communication issues, and limited buffer sizes of a few kilobytes per node. Limited buffers lead to packet drops when overwhelmed.

Main Sources of Congestion

Lecturer: Main sources include channel contention and interference between flows or packets of a single flow, time-variant outgoing channel capacity, high node density increasing contention, and large number of event sources. Events are detected by multiple sources to improve detection efficiency, but this leads to congestion at receiver nodes.

Additional Congestion Sources and Funnel Structure

Lecturer: Closely located sources increase contention, increased flows increase congestion, packet collisions indicate higher congestion, increased reporting rates cause congestion, and the many-to-one inverted funnel-like structure. Multiple sources send data simultaneously toward the sink, creating tapering flows and congestion at intermediate nodes.

Deployment Scenarios and Hotspots

Lecturer: Sensor networks can have densely or sparsely deployed nodes. Sparsely deployed includes low or high data rates. Densely deployed leads to persistent hotspots with ongoing congestion and packet drops.

Hotspot Types in Different Deployments

Lecturer: Sparsely deployed at low rate leads to transient hotspots far from sources, near the sink. Sparsely deployed at high rate leads to both persistent and transient hotspots distributed throughout the network.

Congestion Detection, Notification, and Control

Lecturer: Congestion control involves detection, notification, and control. Detection methods: monitor buffer/queue size, channel busy time and load, interpacket arrival time for data and control packets. Notification to sources via packet header bits or broadcast.

Congestion Control Protocols Overview

Lecturer: Control involves dynamic reporting rate adjustment and in-network data reduction. Protocols include ESRT, CODA, PSFQ, RCRT (focus protocols marked in red).

ESRT Basics and Principles

Lecturer: ESRT: If more packets received than needed, sources reduce sending frequency to reduce congestion and save energy. If too few, sources increase frequency. No end-to-end retransmissions needed. Sink controls flow. Assumptions: trivial network/MAC logic, sinks within network, losses only due to congestion.

ESRT Parameters and Goals

Lecturer: Parameters: Observed event reliability R_i (packets received per decision interval), desired reliability R (application-specific for event detection). Reliability indicator $\eta_i = R_i / R$. Goal: adjust reporting rate so $\eta_i \rightarrow 1$, minimizing energy.

Medium Density Network Plots

Lecturer: Plot of received packets vs. reporting frequency for $n=41, 52, 62$ nodes (medium density). Reliability increases linearly with frequency to a peak, then decreases. Peak shifts left with more nodes (lower max frequency).

High Density Network Plots

Lecturer: High density ($n=81, 90, 101$): similar pattern, increases to $f_{\max}$ then drops drastically. $f_{\max}$ decreases with more nodes. Observations intuitive.

ESRT Network States [

Lecturer: ESRT states: No Congestion Low Reliability (NCLR), No Congestion High Reliability (NCHR), Congestion High Reliability (CHR), Congestion Low/Equal Reliability (CLER), Optimal Operating Region (OOR).

ESRT Actions per State

Lecturer: NCLR: multiplicatively increase f to achieve reliability quickly. NCHR: conservatively decrease f to save energy without compromising reliability. CHR: aggressively decrease f to NCHR, relieve congestion. Formulas in ESRT paper.

ESRT Formulas and Operation

Lecturer: Formulas for f_{i+1} based on states and η_i . Sink handles congestion detection (buffer flags) and reliability calculation per interval. If flag set, high congestion high reliability: reduce reporting rate.

ESRT Concluding Remarks

Lecturer: ESRT reliability application-specific, no individual packet guarantees. Controls via reporting rate changes at sink.

CODA Introduction and Mechanisms

Lecturer: CODA (Congestion Detection and Avoidance) considers energy. Mechanisms: congestion detection, open-loop hop-by-hop backpressure (notify neighbors), closed-loop multi-source regulation (feedback to sources).

CODA Backpressure Example

Lecturer: Open-loop: congested node notifies upstream neighbor (e.g., node 5 to 4, 4 to 1/2), propagating backpressure hop-by-hop toward sources.

CODA Closed-Loop and Detection

Lecturer: Closed-loop: no ACK after congestion (e.g., after packets 4-6), ACK with regulated bit or after recovery (packets 7-8). Detection: buffer occupancy (unreliable in CSMA), channel loading, reporting rate. Combines open (local/transient) and closed (persistent) for control.

CODA Performance Metrics

Lecturer: Metrics: average energy tax (dropped/received packets ratio), average fidelity penalty (vs. ideal scheme). Handles persistent/transient congestion, considers energy unlike ESRT.

Lecture Conclusion

Lecturer: End of first part on congestion and flow control in sensor networks. Thank you.